Corresponding parts of congruent tri[redacted]es

1 Consider the statement,

$$\triangle JET \cong \triangle ZAP$$

State the angle that corresponds to the following:

a $\angle EJT$ **b** $\angle JET$ **c** $\angle JTE$

2 Consider the statement,

$$\triangle ASK \cong \triangle TAX$$

Determine whether the following can be concluded from the statement.

a $\angle SKA \cong \angle ATX$ **b** $\angle AKS \cong \angle TXA$

c $\angle KAS \cong \angle XTA$ **d** $\angle ASK \cong \angle TAX$

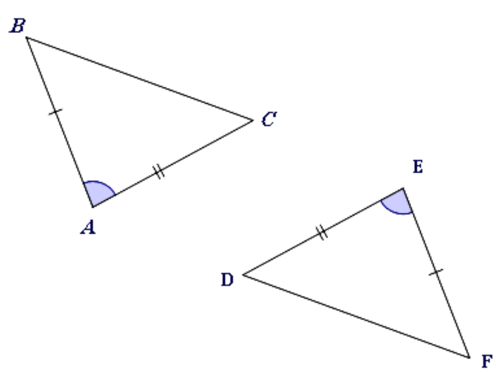
3 For each of the following diagrams:

i State the congruence statement for the triangles.

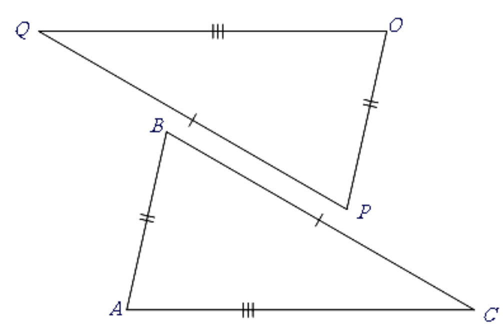
ii State the reason why these two triangles are congruent.

iii Which angle is congruent to $\angle ACB$?

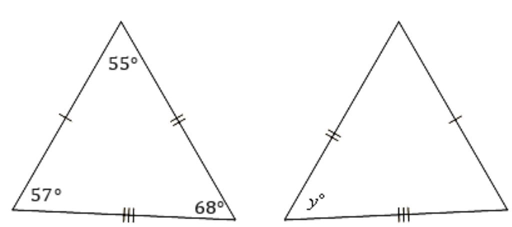
a



b



4 Consider the following diagram:
Find the value of y.

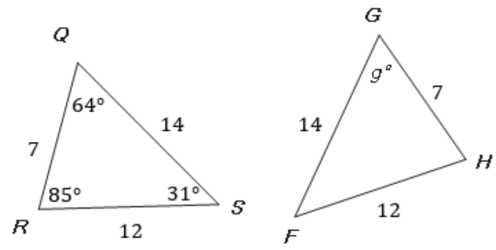


5 Consider the following pair of congruent triangles:

a State the congruence statement for the triangles.

b Identify the congruence postulate that justifies the congruence of the two triangles.

c Hence, solve for g.



6 All corresponding sides and angles of $\triangle RST$ and $\triangle DEF$ are congruent. Determine whether the following statements are always true.

a There is a sequence of transformations that maps $\triangle RST$ to $\triangle DEF$.

b There is a translation followed by a rotation that maps $\overline{RT}$ to $\overline{DF}$.

c There is a dilation that maps $\triangle RST$ to $\triangle DEF$.

d There is a reflection that maps $\overline{RS}$ to $\overline{DE}$.

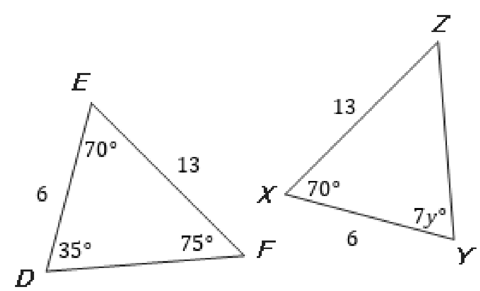
e There is no sequence of rigid motions that maps $\triangle RST$ to $\triangle DEF$.

7 Consider the following diagram, where $m\angle XYZ = 7y^\circ$.

a State the congruence statement for the triangles.

b Identify the congruence postulate that justifies the congruence of the two triangles.

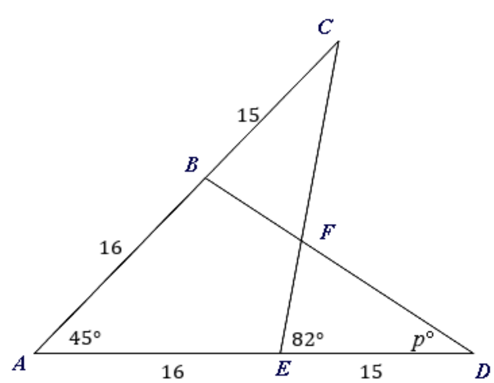
c Hence, solve for y.



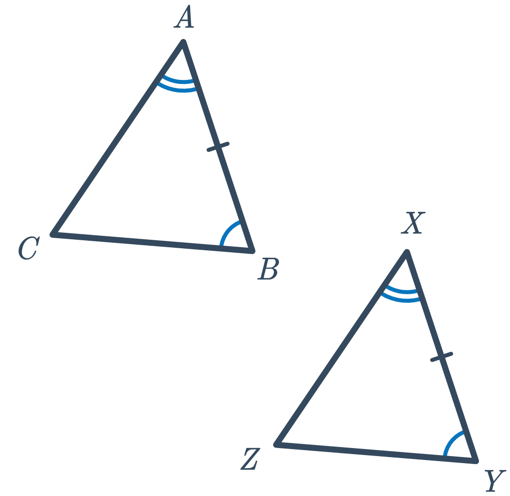
8 Consider the following figure:

a Identify the congruence postulate that justifies $\triangle ACE \cong \triangle ADB$.

b Hence, solve for p.

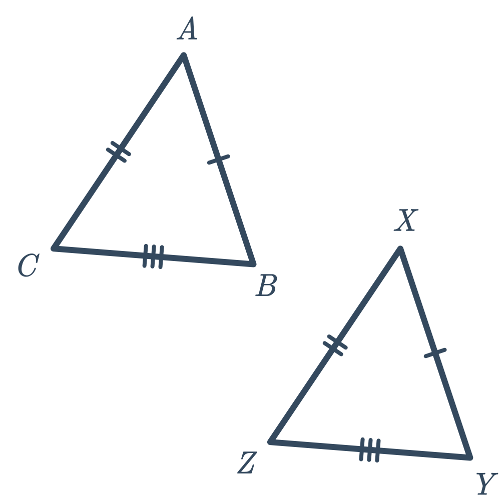


9 Use the information given in the diagram to complete the two-column proof that $\angle ACB \cong \angle XZY$.



	To prove: $\angle ACB \cong \angle XZY$	
	Statements	**Reasons**
1.	$\angle CAB \cong \angle ZXY$	Given
2.	$\angle CBA \cong \angle ZYX$	Given
3.	$\overline{AB} \cong \overline{XY}$	Given
4.	$\triangle ABC \cong \triangle XYZ$	□
5.	$\angle ACB \cong \angle XZY$	Corresponding parts of congruent triangles are congruent (CPCTC)

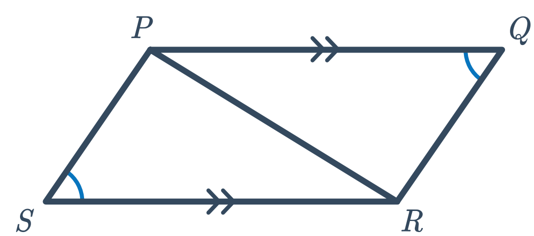
10 Use the information given in the diagram to complete the two-column proof that $\angle ACB \cong \angle XZY$.



To prove: $\angle$[illegible] $\cong \angle XZY$

	Statements	Reasons
1.	$\overline{CB} \cong \overline{ZY}$	Given
2.	$\overline{AC} \cong \overline{XZ}$	Given
3.	$\overline{AB} \cong \overline{XY}$	Given
4.	$\triangle ABC \cong \triangle XYZ$	Side-side-side congruence (SSS)
5.	$\angle ACB \cong \angle XZY$	

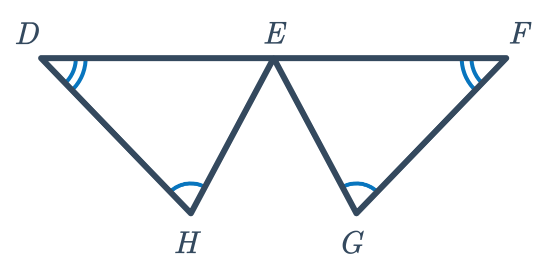
11 Use the information given in the diagram to complete the two-column proof that $\overline{SR} \cong \overline{PQ}$.



To prove: $\overline{SR} \cong \overline{PQ}$

	Statements	Reasons
1.	$\angle PSR \cong \angle RQP$	Given
2.	$\overline{PQ} \parallel \overline{SR}$	Given
3.		
4.	$\overline{PR} \cong \overline{RP}$	Reflexive property of congruence
5.	$\triangle PQR \cong \triangle RSP$	Angle-angle-side congruence (AAS)
6.	$\overline{SP} \cong \overline{RQ}$	

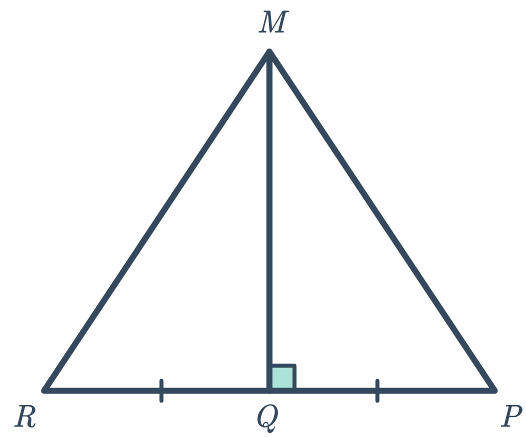
12 Use the information given in the diagram to complete the two-column proof that $\overline{DH} \cong \overline{FG}$.



To prove: $\overline{DH} \cong \overline{FG}$

	Statements	Reasons
1.	E is the midpoint of $\overline{DF}$	Given
2.	$\angle EDH \cong \angle EFG$	Given
3.	$\angle DHE \cong \angle FGE$	Given
4.		
5.	$\triangle DEH \cong \triangle FEG$	Angle-angle-side congruence (AAS)
6.	$\overline{DH} \cong \overline{FG}$	

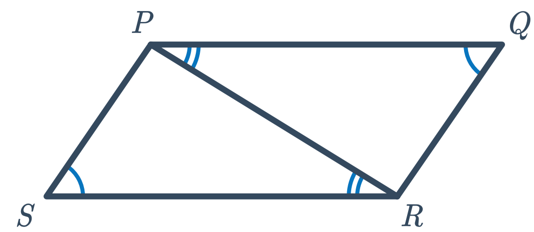
13 Use the information given in the diagram to complete the two-column proof that $\triangle RMP$ is isosceles.



To prove: $\triangle RMP$ is isosceles

	Statements	Reasons
1.	$\overline{RQ} \cong \overline{QP}$	Given
2.	$\overline{RP} \perp \overline{MQ}$	Given
3.	$\angle MQR$ and $\angle MQP$ are right angles	Definition of perpendicular
4.	$\angle MQR \cong \angle MQP$	All right angles are congruent
5.	$\overline{MQ} \cong \overline{MQ}$	Reflexive property of congruence
6.		
7.		
8.	$\triangle RMP$ is isosceles	Isosceles triangles are triangles with two congruent sides

14 Consider the two-column proof which shows that $\overline{SP} \parallel \overline{QR}$ in the following diagram.



To prove: $\overline{SP} \parallel \overline{QR}$

	Statements	Reasons
1.	$\angle PSR \cong \angle RQP$	Given
2.	$\angle SRP \cong \angle QPR$	Given

Determine whether the following correctly proves that $\overline{SP} \parallel \overline{QR}$, if added to the proof.

a

To prove: $\overline{SP} \parallel \overline{QR}$

	Statements	Reasons
1.	$\overline{PR} \cong \overline{PR}$	Reflexive property of congruence
2.	$\triangle SRP \cong \triangle QPR$	Angle-angle-side congruence (AAS)
3.	$\angle SPR \cong \angle QRP$	Corresponding parts of congruent triangles are congruent (CPCTC)
4.	$\overline{SP} \parallel \overline{QR}$	Alternate interior angles converse

b

To prove: $\overline{SP} \parallel \overline{QR}$

	Statements	Reasons
1.	$\overline{SP} \parallel \overline{QR}$	Alternate interior angles converse

c

	To prove: $\overline{SP} \parallel \overline{QR}$	
	Statements	**Reasons**
1.	$\overline{PR} \cong \overline{PR}$	Reflexive property of congruence
2.	$\triangle SRP \cong \triangle QPR$	Angle-angle-side congruence (AAS)
3.	$\overline{SP} \parallel \overline{QR}$	Corresponding parts of congruent triangles are congruent (CPCTC)

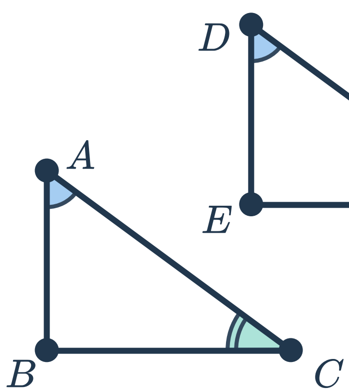
d

	To prove: $\overline{SP} \parallel \overline{QR}$	
	Statements	**Reasons**
1.	$\angle SPR \cong \angle QRP$	Third angles theorem
2.	$\overline{SP} \parallel \overline{QR}$	Alternate interior angles converse

Additional questions

15 State the side corresponding to $\overline{AB}$ in each pair of congruent shapes:

a



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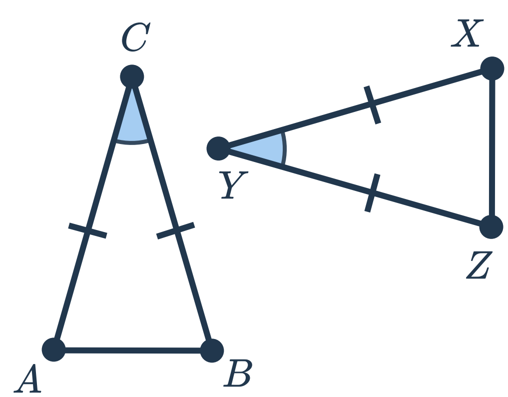
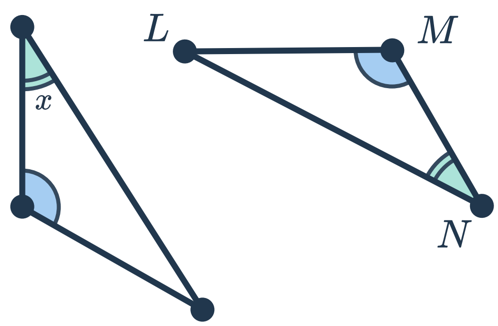


16 State the angle corresponding to the angle [redacted] measure of x in each pair of congruent shapes:

a



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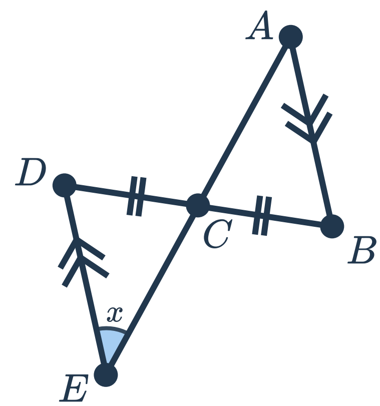

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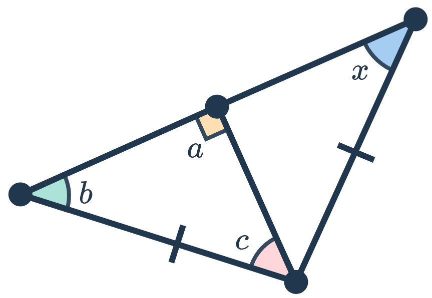


17 Consider the figure:

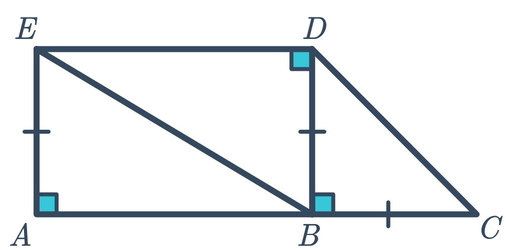
a Find the triangle congruent to $\triangle EAB$.

b Find the angle that corresponds to:

i $\angle EBA$ ii $\angle BEA$

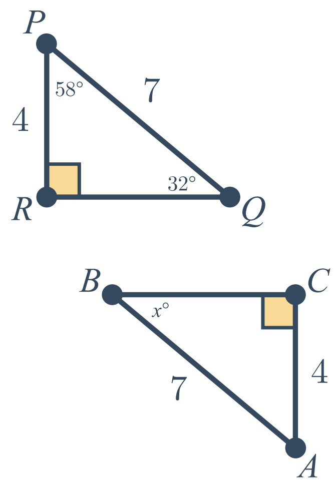
c Find the side that corresponds to:

i $\overline{AB}$ ii $\overline{AE}$

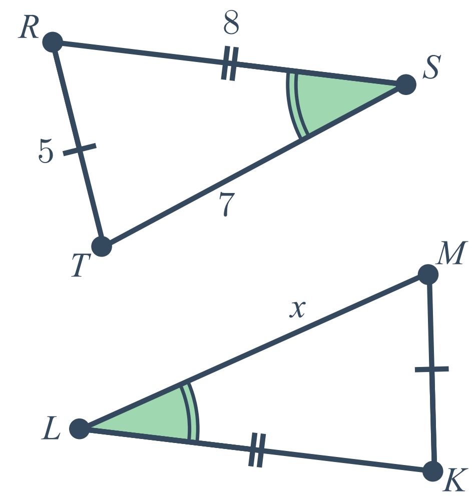


18 The triangles shown in the diagrams are congruent. Find the value of x.

a

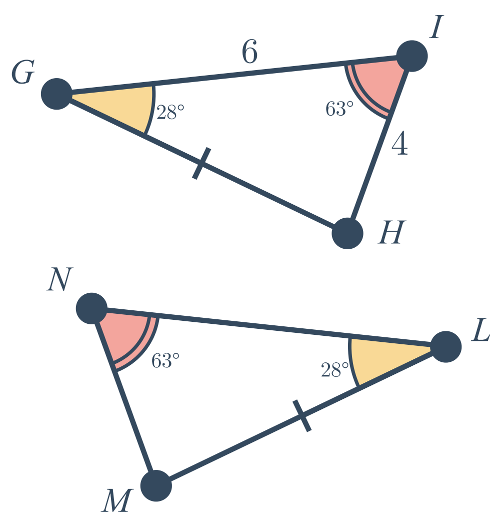


b

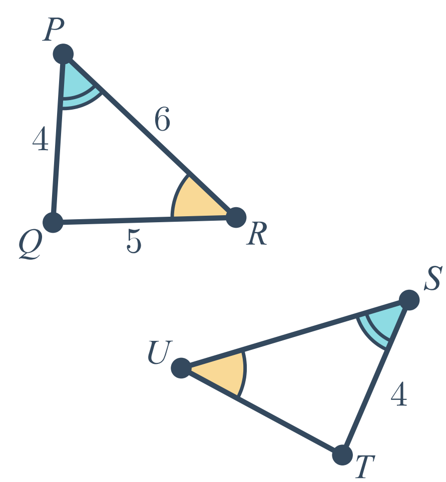


19 The triangles shown in the diagrams are congruent.

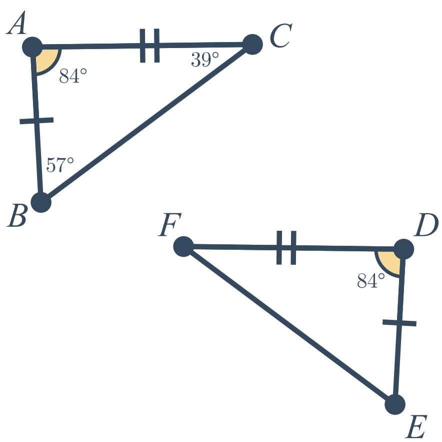
a Find NM.

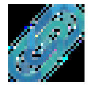


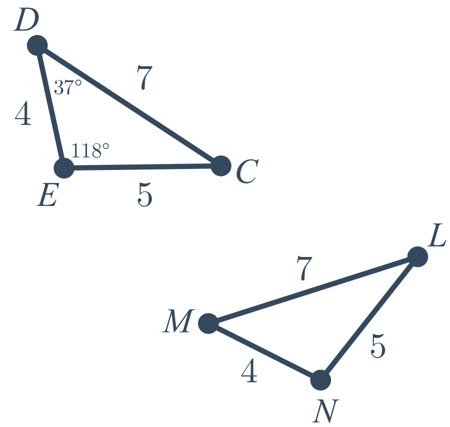
b Find SU.



c Find $m\angle DEF$.



 Find$m\angle NLM$.



20

a Write two different congruence statements using the points D, E, F.
$\Delta ABC \cong$ ▯ $\Delta ABC \cong$ ▯

b List all corresponding sides and angles for each congruency statement.

c Explain why the order of the points in the congruence statement is important.

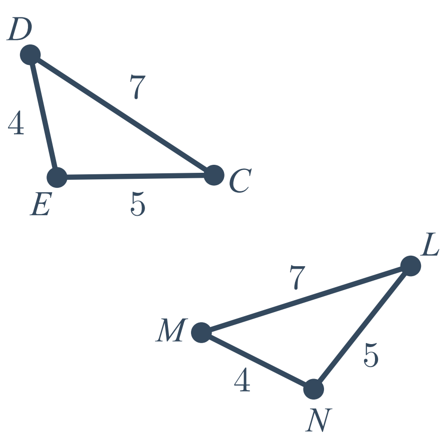
21

i Identify the congruence postulate justifies the congruence of the triangles shown in the diagram.

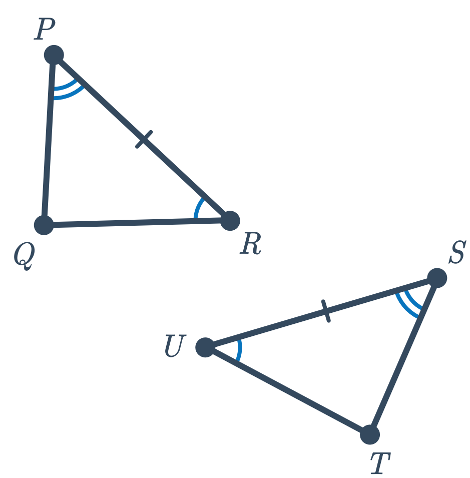
ii Write a congruence statement for the triangles.

iii Identify all additional pairs of congruent parts using the fact that corresponding parts of congruent triangles are congruent (CPCTC).

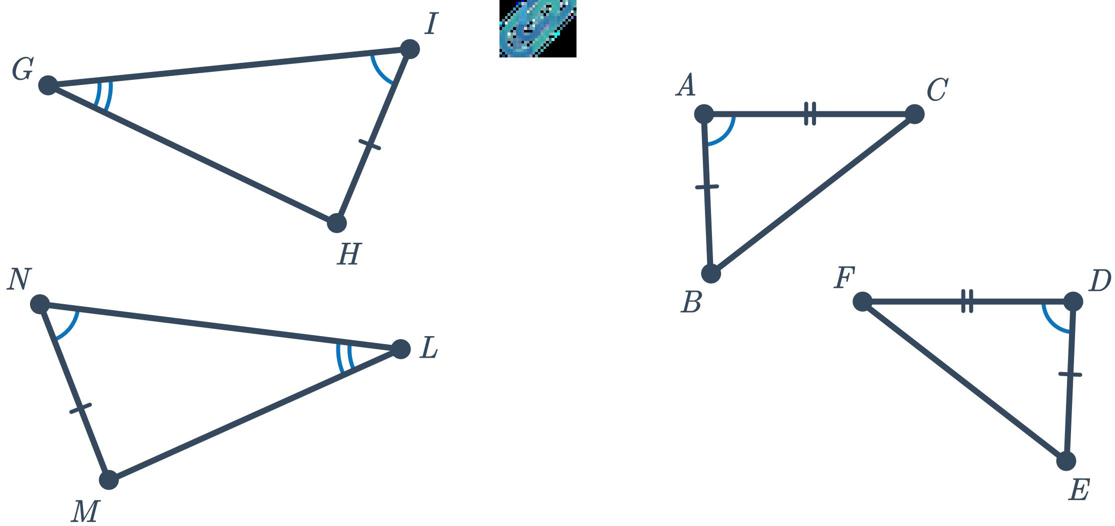
a



b



c

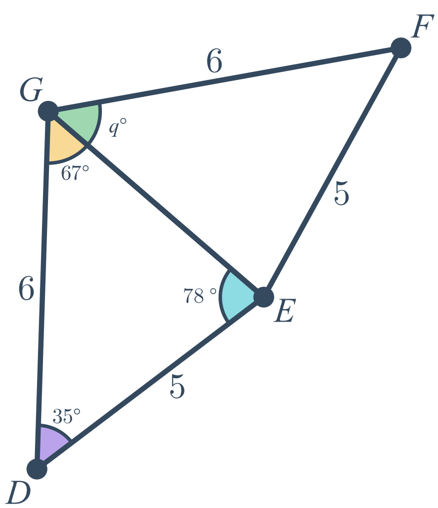


22 For each of the following diagrams:

i Identify the congruence postulate that justifies the congruence of the triangles shown in the diagram.

ii Find the value of the variable.

a



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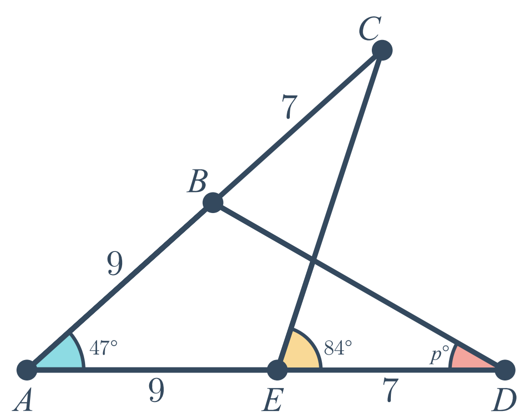
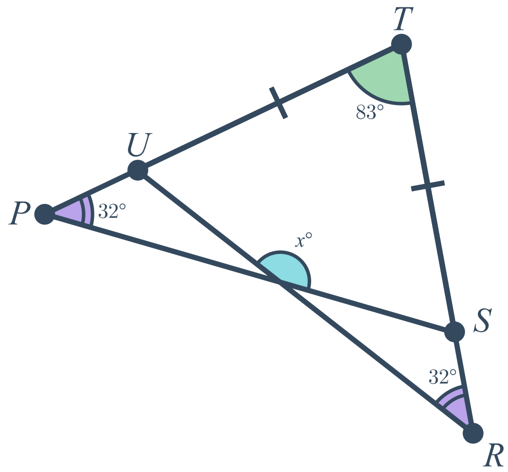
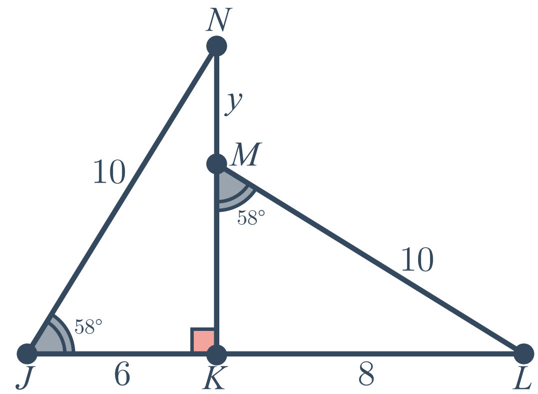


c

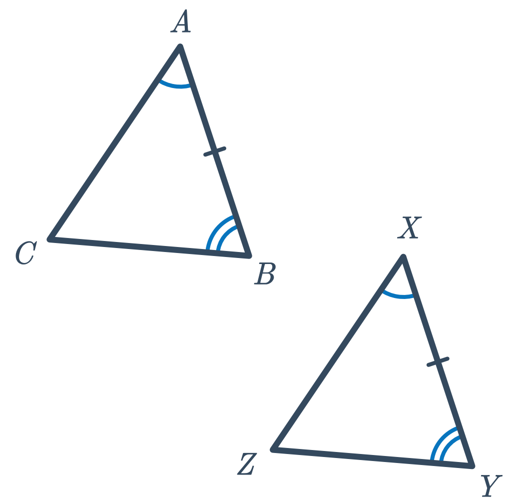




23 Given:

- $\angle A \cong \angle X$
- $\angle B \cong \angle Y$
- $\overline{AB} \cong \overline{XY}$

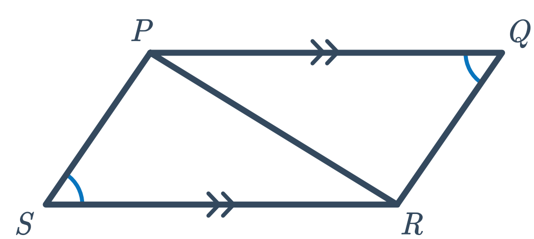
Prove: $\overline{BC} \cong \overline{YZ}$



24 Given:

- $\overline{PQ} \parallel \overline{RS}$
- $\angle S \cong \angle Q$

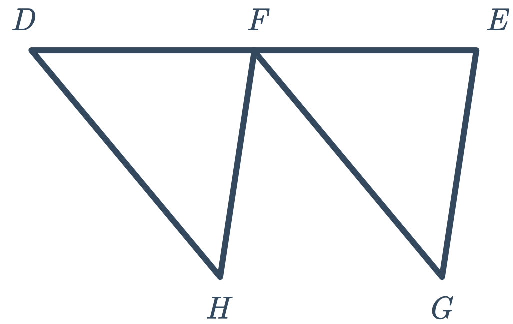
Prove: $\overline{PS} \cong \overline{QR}$



25 Given:

- F is the midpoint of $\overline{DE}$
- $\overline{DH} \parallel \overline{FG}$
- $\overline{DH} \cong \overline{FG}$

Prove: $\overline{FH} \parallel \overline{EG}$



26 Use the information given in the diagram to prove $\triangle PQX \cong \triangle RQX$.